



The tap was right. It landed too early.

Why a test that does everything right still flakes on mobile, and how Mobium asks the app when it is done. Measured on an emulator, a simulator, a Pixel 8 Pro and an iPhone 15 Plus.

row B, generation 1: **stale** launched normally

row B, generation 2: **current** with `--gray-box`



Every wait passed. The tap still missed.

```
$ mobium tap testid=busyQuiet  
$ mobium tap testid=busyRowB  
$ mobium text testid=busyOutcome  
row B, generation 1: stale
```

Before it taps, Mobium waits for the target to pass four checks. Row B passed all four the whole time:

on screen PASSED

still PASSED

enabled PASSED

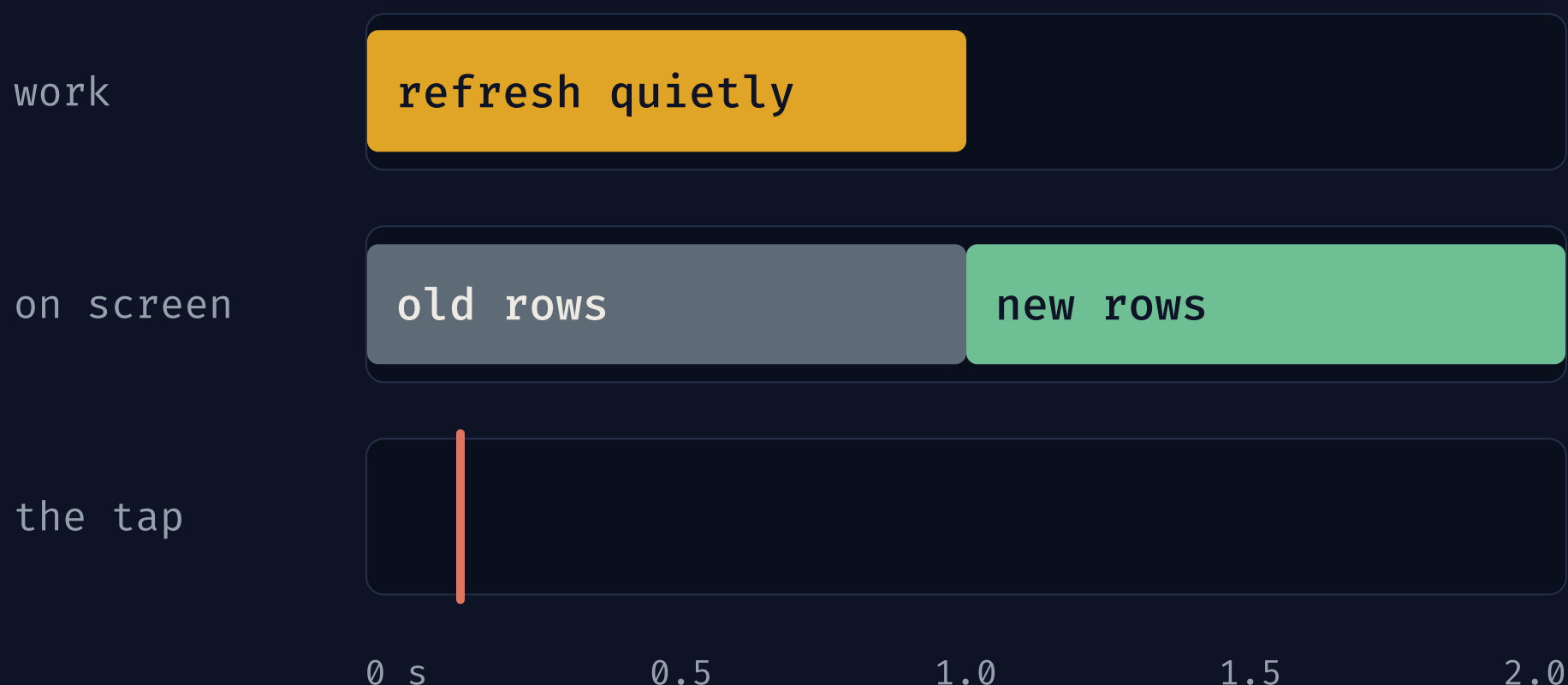
uncovered PASSED

It was also about to be replaced, and nothing on screen said so.



Only the app knows it is still busy.

MobiumApp's Busy Demo starts 0.4 to 1.6 seconds of work and leaves the old rows up while it runs. A tap that comes before the new rows lands on a row that is about to go.



On a real Pixel and iPhone, 7 to 9 taps in 10 landed stale.



So the app says when it is busy.

```
const refresh = () => {  
  GrayBox.busy('quiet');  
  // the work, then the new rows  
};  
// once the new rows have rendered:  
GrayBox.idle('quiet');
```

MobiumApp links Mobium's gray-box library, in Swift and Kotlin. It writes one line to the device log for each change, and Mobium already reads that log:

```
MOBIUM-GRAYBOX busy=1 tag=quiet  
MOBIUM-GRAYBOX still busy=1  
MOBIUM-GRAYBOX busy=0 tag=quiet
```



One flag. The tap waits for the app.

```
$ mobium launch --gray-box dev.mobium.mobiumapp
$ mobium tap testid=busyRowB
gray box: waited 955 ms for the app to go idle
(busy: quiet)
```

LAUNCHED NORMALLY

Back

Busy

row B, generation 1: stale

Both refresh the rows after 0.4 to 1.6 s. Refresh shows a spinner meanwhile; Refresh quietly leaves the old rows up.

Refresh

Refresh quietly

Refresh from an alert

Refresh twice

Keep polling

WITH --GRAY-BOX

Back

Busy

row B, generation 2: current

Both refresh the rows after 0.4 to 1.6 s. Refresh shows a spinner meanwhile; Refresh quietly leaves the old rows up.

Refresh

Refresh quietly

Refresh from an alert

Refresh twice

Keep polling



The same test, one key apart.

```
"app": "dev.mobium.mobiumapp",  
+ "grayBox": true,
```

Five runs each way, on four devices:

DEVICE	WITHOUT	WITH	WAITED
Pixel 7 emulator	0 of 5	5 of 5	1148 ms
iPhone 17 Pro simulator	0 of 5	5 of 5	1202 ms
Pixel 8 Pro	3 of 5	5 of 5	709 ms
iPhone 15 Plus	0 of 5	5 of 5	955 ms

Passing 3 of 5 is what flaky looks like: it depends on how long the work happened to take.



A wait that knows when to stop.

Work that never ends

Refused after 10 seconds, naming what kept the app busy.

The app goes home, or crashes

Not waited on. Busy is a lease the app renews twice a second, and a dead app stops renewing it.

The log stream drops

The result says Mobium could not hear the app, and it listens again.

Every result says what it waited for

“waited 955 ms (busy: quiet)”, “the app was idle”, or why it did not wait.

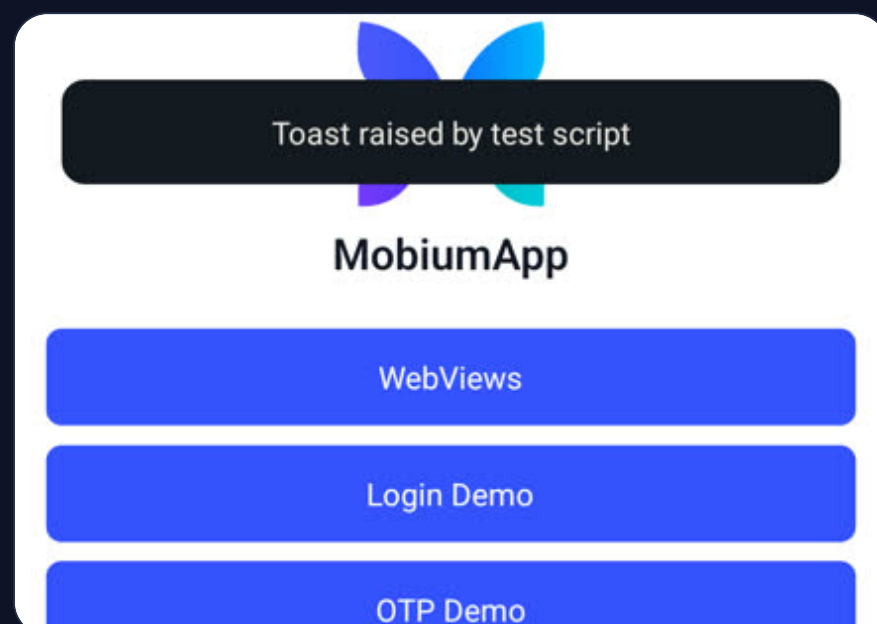
All six edge checks pass on all four devices.



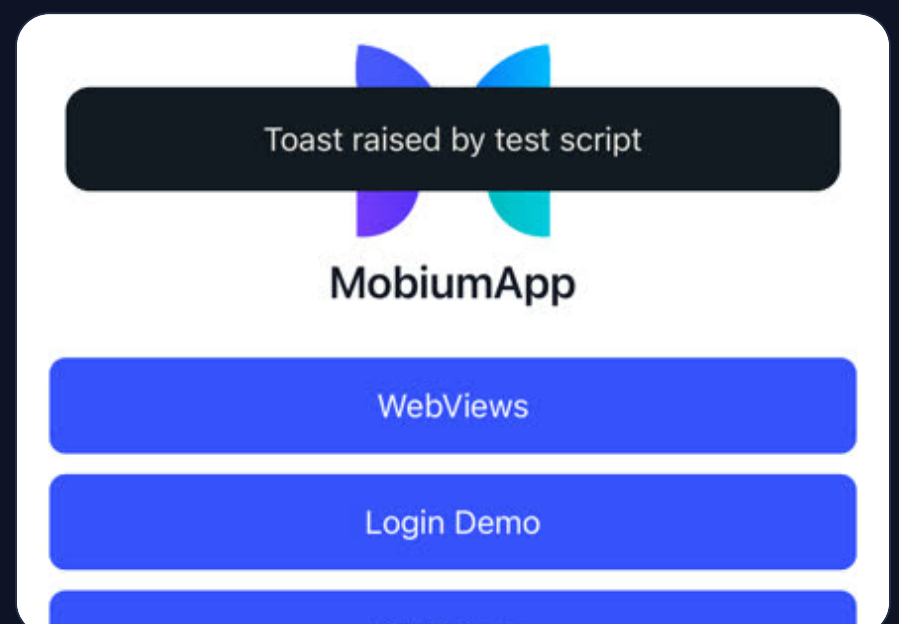
The test asks the app to do something.

```
// the app, in a build made for testing
GrayBox.register("raiseToast", (message) => {
  ToastAndroid.show(message, ToastAndroid.SHORT);
  return "shown";
});
# the test
device.hook("raiseToast", "Toast raised by test
script") # 'shown'
```

PIXEL 8 PRO



IPHONE 15 PLUS



sign in, seed data, raise a toast

A door the app opens, and only for the test.

The app opts in

It links a small library and says what it is doing. Work it does not declare is not waited for.

Silent unless the test turns it on

A launch argument on iOS, an intent extra on Android. Anyone opening the same build sees nothing.

Only what the app registered

A hook is called by name. Any other name is refused, with the list of names that exist.

Builds made for testing

Keep hooks out of what ships. An app without the library is driven the ordinary way.





See the race, then fix it, in a few minutes.

```
$ go install  
github.com/mobiumdev/mobium/cmd/mobium@latest  
$ mobium launch --gray-box dev.mobium.mobiumapp
```

MOBIUM
github.com/mobiumdev/mobium

QUICK START, TUTORIAL, EVIDENCE
github.com/lana-20/mobium-graybox-demo

THE TALK, WITH EVERY DEVICE'S RECORDINGS
lana-20.github.io/mobium-graybox-demo

